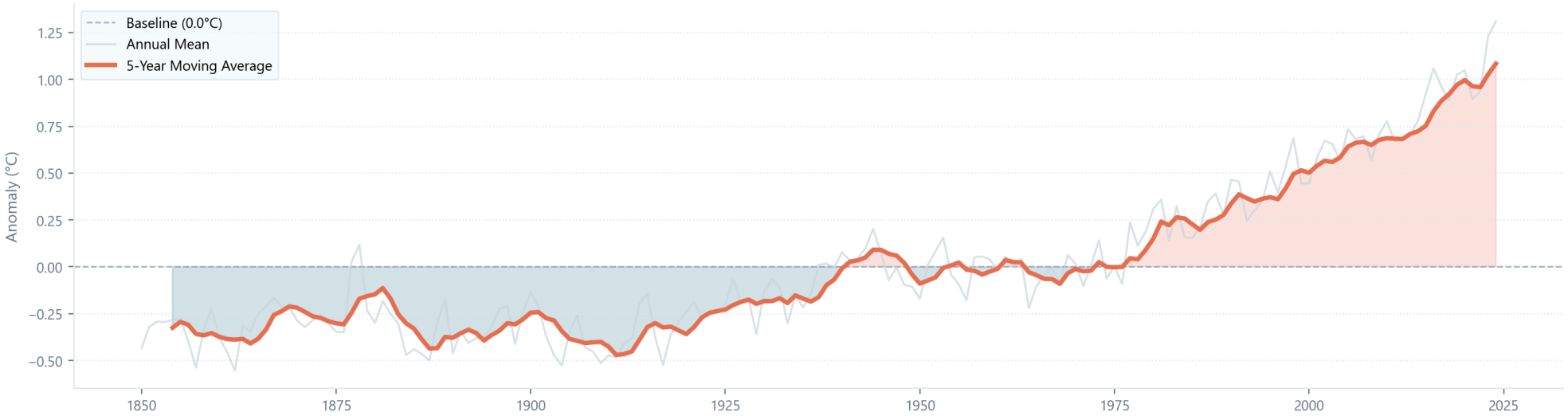
CLIMATE CHANGE ANALYSIS

Global Monthly Temperature Anomalies (1850 – 2024) • Environmental Intelligence Dashboard

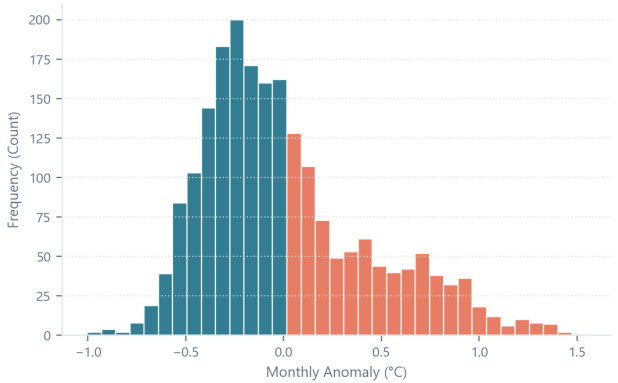
TIMELINE FILTER
1850 – 2024 (175 Years)

AVERAGE ANOMALY	HIGHEST ANOMALY	LOWEST ANOMALY	MONTHLY RECORDS
+0.01 °C	**+1.55 °C**	**-1.00 °C**	**2,100**
Baseline: 1951–1980 Mean	Recorded: September 2023	Recorded: January 1861	100% Complete • Cleaned

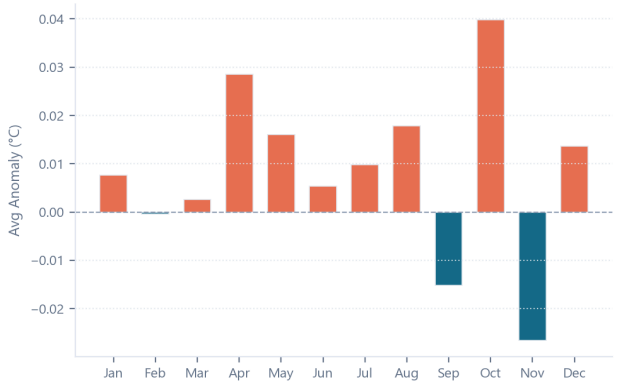
Global Temperature Anomaly Trend (1850 – 2024)



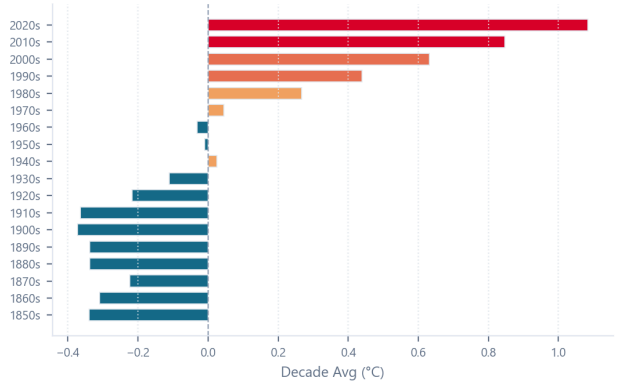
Distribution of Monthly Anomalies



Average Anomaly by Month



Average Anomaly by Decade



ANALYTICAL FINDINGS & CLIMATE TRENDS

Decadal Warming Progression, Extreme Temperature Events & Statistical Distribution

ANALYSIS TYPE
Decadal & Seasonal Dynamics

Decade-by-Decade Warming Progression

Tracking the average monthly anomaly (°C) grouped by decade demonstrates a distinct phase shift starting in the late 20th century:

Decade	Avg Anomaly	Variance vs Base	Climate Regime
1850s	-0.340 °C	-0.34 °C	Pre-Industrial Base
1880s	-0.338 °C	-0.34 °C	Early Industrial
1910s	-0.364 °C	-0.36 °C	Cool Baseline
1940s	+0.025 °C	+0.03 °C	Neutral Period
1970s	+0.045 °C	+0.05 °C	Slight Warming
1980s	+0.267 °C	+0.27 °C	Rapid Rise Phase
1990s	+0.439 °C	+0.44 °C	Accelerated Heat
2000s	+0.631 °C	+0.63 °C	Strong Warming
2010s	+0.847 °C	+0.85 °C	Record Decadal High
2020s (to 2024)	**+1.084 °C**	**+1.08 °C**	Extreme Heat Era

Executive Insights & Risk Implications

- **Sharp Rate of Acceleration:** Global temperatures remained steady within -0.4°C to 0.0°C from 1850 through the 1970s. Since 1980, the rate of increase surged to approximately +0.18°C per decade.
- **Historic Records:** The highest recorded single-month anomaly in recorded history occurred in **September 2023 (+1.55°C)**, closely followed by continuous +1.2°C to +1.5°C anomalies throughout 2024.
- **Seasonal Symmetry:** Analyzing monthly variations across all 175 years reveals that global temperature anomalies are consistently distributed across all 12 calendar months (varying only by ±0.03°C on average), demonstrating that global warming is a planetary continuous phenomenon rather than a seasonal quirk.
- **Distribution Skewness:** The histogram exhibits a strong right-hand positive skew, confirming that positive temperature extremes now outnumber cold baseline deviations by a substantial margin.

Recommended Decision Actions

For corporate ESG compliance, municipal urban resilience planning, and energy demand forecasting, projections must adopt the 2010–2024 trajectory rather than historic 100-year averages.

TECHNICAL ARCHITECTURE & DAX SPECIFICATION

Data Model Schema, ETL Pipeline, DAX Formula Definitions & Power BI PBIP Setup

PROJECT SPEC
PBIP / Fabric PBIR Format

Data Pipeline & Preprocessing

Data pipeline implemented via Power Query (M) and automated ETL:

- **Deduplication:** The raw input dataset contained two identical blocks (4,200 records). Deduplicated to 2,100 unique Year-Month pairs (1850-01 to 2024-12).
- **Null Integrity:** Blank measurements preserved as null rather than defaulted to zero to prevent statistical distortion.
- **Calculated Columns:** Added `Decade = Number.RoundDown([Year] / 10) * 10` and 0.1°C histogram bins for distribution clustering.

Power BI Semantic Model Schema

Column Name	Data Type	Role
Year	Int64	Temporal Dimension (1850–2024)
Month	Int64	Monthly Dimension (1–12)
Date	String / Date	Continuous Time Axis
Monthly Anomaly (°C)	Double	Primary Temperature Metric
Monthly Uncertainty (°C)	Double	Confidence Interval (±95%)
Decade	String	Decadal Aggregation Group

DAX Measure Definitions

1. Average Monthly Anomaly (°C)

```
Average Monthly Anomaly = AVERAGE('Cleaned_Temperature_Data'[Monthly Anomaly (°C)])
```

2. Highest Monthly Anomaly (°C)

```
Highest Monthly Anomaly = MAX('Cleaned_Temperature_Data'[Monthly Anomaly (°C)])
```

3. Lowest Monthly Anomaly (°C)

```
Lowest Monthly Anomaly = MIN('Cleaned_Temperature_Data'[Monthly Anomaly (°C)])
```

4. Total Monthly Records

```
Monthly Records = COUNT('Cleaned_Temperature_Data'[Monthly Anomaly (°C)])
```

5. Annual Average Anomaly

```
Annual Average Anomaly = AVERAGEX(VALUES('Cleaned_Temperature_Data'[Year]),
CALCULATE(AVERAGE('Cleaned_Temperature_Data'[Monthly Anomaly (°C)])))
```